MSG No. 108.238 I_c4cm [Type IV, tetragonal]

Table 1: Wyckoff site: `4a`, site symmetry: `4m'm'`

No.	position	mapping
1	$[0,\ 0,\ z]$	`[1,2,3,4,21,22,23,24]`
2	$\left[0,\ 0,\ z+\frac{1}{2}\right]$	`[5,6,7,8,17,18,19,20]`
3	$\left[\frac{1}{2},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[9,10,11,12,29,30,31,32]`
4	$\left[\frac{1}{2},\ \frac{1}{2},\ z\right]$	`[13,14,15,16,25,26,27,28]`

Table 2: Wyckoff site: `4b`, site symmetry: `4'm'm`

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ z\right]$	`[1,4,15,16,21,22,26,27]`
2	$\left[\frac{1}{2},\ 0,\ z\right]$	`[2,3,13,14,23,24,25,28]`
3	$\left[0,\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[5,6,10,11,17,20,31,32]`
4	$\left[\frac{1}{2},\ 0,\ z+\frac{1}{2}\right]$	`[7,8,9,12,18,19,29,30]`

Table 3: Wyckoff site: `8c`, site symmetry: `2.mm`

No.	position	mapping
1	$\left[\frac{1}{4},\ \frac{1}{4},\ z\right]$	`[1,15,24,28]`
2	$\left[\frac{3}{4},\ \frac{1}{4},\ z\right]$	`[2,14,21,27]`
3	$\left[\frac{1}{4},\ \frac{3}{4},\ z\right]$	`[3,13,22,26]`
4	$\left[\frac{3}{4},\ \frac{3}{4},\ z\right]$	`[4,16,23,25]`
5	$\left[\frac{3}{4},\ \frac{1}{4},\ z+\frac{1}{2}\right]$	`[5,11,18,30]`
6	$\left[\frac{1}{4},\ \frac{3}{4},\ z+\frac{1}{2}\right]$	`[6,10,19,29]`
7	$\left[\frac{3}{4},\ \frac{3}{4},\ z+\frac{1}{2}\right]$	`[7,9,20,32]`
8	$\left[\frac{1}{4},\ \frac{1}{4},\ z+\frac{1}{2}\right]$	`[8,12,17,31]`

Table 4: Wyckoff site: `16d`, site symmetry: `.m'.`

No.	position	mapping
1	$[0,\ y,\ z]$	`[1,21]`
2	$[-y,\ 0,\ z]$	`[2,23]`
3	$[y,\ 0,\ z]$	`[3,24]`
4	$[0,\ -y,\ z]$	`[4,22]`
5	$\left[0,\ y,\ z+\frac{1}{2}\right]$	`[5,17]`
6	$\left[0,\ -y,\ z+\frac{1}{2}\right]$	`[6,20]`
7	$\left[-y,\ 0,\ z+\frac{1}{2}\right]$	`[7,18]`
8	$\left[y,\ 0,\ z+\frac{1}{2}\right]$	`[8,19]`
9	$\left[\frac{1}{2},\ y+\frac{1}{2},\ z+\frac{1}{2}\right]$	`[9,29]`
10	$\left[\frac{1}{2}-y,\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[10,31]`
11	$\left[y+\frac{1}{2},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[11,32]`
12	$\left[\frac{1}{2},\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$	`[12,30]`
13	$\left[\frac{1}{2},\ y+\frac{1}{2},\ z\right]$	`[13,25]`

continued ...

Table 4

No.	position	mapping
14	$\left[\frac{1}{2},\ \frac{1}{2}-y,\ z\right]$	`[14,28]`
15	$\left[\frac{1}{2}-y,\ \frac{1}{2},\ z\right]$	`[15,26]`
16	$\left[y+\frac{1}{2},\ \frac{1}{2},\ z\right]$	`[16,27]`

Table 5: Wyckoff site: `16e`, site symmetry: `..m'`

No.	position	mapping
1	$[x,\ x,\ z]$	`[1,24]`
2	$[-x,\ x,\ z]$	`[2,21]`
3	$[x,\ -x,\ z]$	`[3,22]`
4	$[-x,\ -x,\ z]$	`[4,23]`
5	$\left[-x,\ x,\ z+\frac{1}{2}\right]$	`[5,18]`
6	$\left[x,\ -x,\ z+\frac{1}{2}\right]$	`[6,19]`
7	$\left[-x,\ -x,\ z+\frac{1}{2}\right]$	`[7,20]`
8	$\left[x,\ x,\ z+\frac{1}{2}\right]$	`[8,17]`
9	$\left[x+\frac{1}{2},\ x+\frac{1}{2},\ z+\frac{1}{2}\right]$	`[9,32]`
10	$\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ z+\frac{1}{2}\right]$	`[10,29]`
11	$\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ z+\frac{1}{2}\right]$	`[11,30]`
12	$\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ z+\frac{1}{2}\right]$	`[12,31]`
13	$\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ z\right]$	`[13,26]`
14	$\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ z\right]$	`[14,27]`
15	$\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ z\right]$	`[15,28]`
16	$\left[x+\frac{1}{2},\ x+\frac{1}{2},\ z\right]$	`[16,25]`

Table 6: Wyckoff site: `16f`, site symmetry: `..m`

No.	position	mapping
1	$\left[x,\ x+\frac{1}{2},\ z\right]$	`[1,16]`
2	$\left[\frac{1}{2}-x,\ x,\ z\right]$	`[2,13]`
3	$\left[x+\frac{1}{2},\ -x,\ z\right]$	`[3,14]`
4	$\left[-x,\ \frac{1}{2}-x,\ z\right]$	`[4,15]`
5	$\left[-x,\ x+\frac{1}{2},\ z+\frac{1}{2}\right]$	`[5,10]`
6	$\left[x,\ \frac{1}{2}-x,\ z+\frac{1}{2}\right]$	`[6,11]`
7	$\left[\frac{1}{2}-x,\ -x,\ z+\frac{1}{2}\right]$	`[7,12]`
8	$\left[x+\frac{1}{2},\ x,\ z+\frac{1}{2}\right]$	`[8,9]`
9	$\left[x,\ x+\frac{1}{2},\ z+\frac{1}{2}\right]$	`[17,32]`
10	$\left[\frac{1}{2}-x,\ x,\ z+\frac{1}{2}\right]$	`[18,29]`
11	$\left[x+\frac{1}{2},\ -x,\ z+\frac{1}{2}\right]$	`[19,30]`
12	$\left[-x,\ \frac{1}{2}-x,\ z+\frac{1}{2}\right]$	`[20,31]`
13	$\left[-x,\ x+\frac{1}{2},\ z\right]$	`[21,26]`
14	$\left[x,\ \frac{1}{2}-x,\ z\right]$	`[22,27]`
15	$\left[\frac{1}{2}-x,\ -x,\ z\right]$	`[23,28]`
16	$\left[x+\frac{1}{2},\ x,\ z\right]$	`[24,25]`

Table 7: Wyckoff site: 32g, site symmetry: 1

No.	position	mapping
1	$[x,\ y,\ z]$	[1]
2	$[-y,\ x,\ z]$	[2]
3	$[y,\ -x,\ z]$	[3]
4	$[-x,\ -y,\ z]$	[4]
5	$\left[-x,\ y,\ z+\frac{1}{2}\right]$	[5]
6	$\left[x,\ -y,\ z+\frac{1}{2}\right]$	[6]
7	$\left[-y,\ -x,\ z+\frac{1}{2}\right]$	[7]
8	$\left[y,\ x,\ z+\frac{1}{2}\right]$	[8]
9	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ z+\frac{1}{2}\right]$	[9]
10	$\left[\frac{1}{2}-y,\ x+\frac{1}{2},\ z+\frac{1}{2}\right]$	[10]
11	$\left[y+\frac{1}{2},\ \frac{1}{2}-x,\ z+\frac{1}{2}\right]$	[11]
12	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$	[12]
13	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ z\right]$	[13]
14	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ z\right]$	[14]
15	$\left[\frac{1}{2}-y,\ \frac{1}{2}-x,\ z\right]$	[15]
16	$\left[y+\frac{1}{2},\ x+\frac{1}{2},\ z\right]$	[16]
17	$\left[x,\ y,\ z+\frac{1}{2}\right]$	[17]
18	$\left[-y,\ x,\ z+\frac{1}{2}\right]$	[18]
19	$\left[y,\ -x,\ z+\frac{1}{2}\right]$	[19]
20	$\left[-x,\ -y,\ z+\frac{1}{2}\right]$	[20]
21	$[-x,\ y,\ z]$	[21]
22	$[x,\ -y,\ z]$	[22]
23	$[-y,\ -x,\ z]$	[23]
24	$[y,\ x,\ z]$	[24]
25	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ z\right]$	[25]
26	$\left[\frac{1}{2}-y,\ x+\frac{1}{2},\ z\right]$	[26]
27	$\left[y+\frac{1}{2},\ \frac{1}{2}-x,\ z\right]$	[27]
28	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ z\right]$	[28]
29	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ z+\frac{1}{2}\right]$	[29]
30	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$	[30]
31	$\left[\frac{1}{2}-y,\ \frac{1}{2}-x,\ z+\frac{1}{2}\right]$	[31]
32	$\left[y+\frac{1}{2},\ x+\frac{1}{2},\ z+\frac{1}{2}\right]$	[32]